

Huy Le

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EDUCATION

University of Pittsburgh

Doctor of Philosophy in Statistics

Dissertation defended; Advisor: Chris McKenna, PhD

Pittsburgh, PA

Expected August 2026

University of California, Berkeley

Master of Arts in Statistics

Berkeley, CA

2019

Oregon State University

Bachelor of Science in Mathematics, Minor in Actuarial Science

GPA: 4.0/4.0, Summa Cum Laude, Award for Outstanding Performance in Mathematics, Oregon State University

Corvallis, OR

2018

SKILLS

Programming: Python, R, SQL, MATLAB, Git, Bash/Shell

Methods: Probability, Stochastic Processes, Statistical Inference, Time Series, Optimization, Bayesian Modeling, Empirical Bayes, Gaussian Processes, Simulation Studies, Model Validation, Machine Learning

RESEARCH EXPERIENCE

Graduate Research Assistant

Department of Statistics, University of Pittsburgh

June 2020 – Present

Pittsburgh, PA

Bayesian Spatiotemporal Modeling for Longitudinal High-Dimensional Data

- Developed Bayesian spatiotemporal models in R and Python for high-dimensional longitudinal data, combining latent Gaussian processes, spike-and-slab priors, and covariance estimation to identify persistent and time-varying signals.
- Built an end-to-end inference pipeline using EM-style estimation, posterior simulation, and region-level error-control procedures for statistically calibrated signal detection.
- Designed simulation studies to evaluate calibration, power, false discovery control, and robustness under varying signal, noise, dependence, and heterogeneity regimes.
- Applied the framework to longitudinal cohort data with 750,000+ ordered features, translating feature-level summaries into interpretable region-level inference.

Spatial Empirical Bayes Modeling for Heterogeneous Correlated Data

- Developed empirical Bayes models for heterogeneous spatial dependence, allowing local inference to adapt when correlation structure varies across ordered features.
- Implemented scalable approximation and optimization methods, including expectation propagation, importance sampling, pseudo-likelihood estimation, and mixture optimization.
- Evaluated model behavior against standard region-detection baselines through simulation studies focused on calibration, localization, and posterior uncertainty quantification.
- Produced posterior summaries, diagnostic checks, and reproducible analyses to support statistically grounded interpretation of high-dimensional correlated data.

PROFESSIONAL EXPERIENCE

Quantitative Modeling Intern (PK/PD)

AbbVie

June 2023 – August 2023

South San Francisco, CA

- Developed and validated 6 ODE-based models in MATLAB to support drug feasibility and development-stage decision making.
- Built and deployed R Shiny simulation applications for real-time exploration of concentration–response relationships.
- Compared model predictions against experimental data to assess predictive performance and support candidate evaluation.
- Implemented Git-based version control to improve reproducibility and collaboration across modeling workflows.

Teaching Assistant and Instructor

Department of Statistics, University of Pittsburgh

June 2020 – Present

Pittsburgh, PA

- Primary instructor or teaching assistant for probability, mathematical statistics, and introductory/intermediate statistics courses; communicated technical ideas to students with varied quantitative backgrounds.

PROJECTS

Spatial Risk Forecasting and Route Optimization

University of California, Berkeley

January 2019 – May 2019

Berkeley, CA

- Built a crime-risk forecasting model in R using zero-inflated negative binomial regression for overdispersed, zero-inflated count data.
- Designed a route-optimization algorithm that combined spatial risk forecasts with public safety data to identify statistically safer routes.